

Dr. Eng., Nur Hamid, S.Si., M.Kom.



Current Affiliation: Interdisciplinary Research Center for Smart Mobility and Logistics (IRC-SML), Building 23, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia

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I. HIGHLIGHTS

- Currently a postdoctoral fellow at the Interdisciplinary Research Center for Smart Mobility and Logistics (IRC-SML), King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia. Start joining the research center in January 2025.
- Research interests: optimization algorithm, machine learning, deep learning, reinforcement learning, image processing, 3D simulation, mobile robotics, health data processing.

II. EDUCATION

Doctoral Degree 2021 – 2024	Field: Electrical Engineering and Computer Science Institution: Kanazawa University (https://www.kanazawa-u.ac.jp/en/) Result: CGPA 3.83 out of 4.00 scale Doctoral degree dissertation: Nature-Inspired Optimization Algorithms for Unmanned Surface Vehicle (USV) in Complex Environment Supervisor: Prof. Hidetaka Nambo.
Master's Degree 2018 – 2020	Field: Computer Science Institution: Universitas Indonesia (https://cs.ui.ac.id/) Result: CGPA 3.62 out of 4.00 scale Master's degree Thesis: Adaptive Update in Dynamic Graph Convolutional Neural Network for land cover semantic segmentation of LiDAR data Supervisor: Prof. Dr. Eng. Wisnu Jatmiko.
Bachelor's degree 2011 – 2015	Field: Physics (Electronic & Instrumentation) Institution: Diponegoro University (https://fisika.fsm.undip.ac.id/) Result: CGPA 3.82 out of 4.00 scale Undergraduate Thesis: Determination of color blindness level using fuzzy color space segmentation methods and rule-based forward chaining on Ishihara images Supervisor: Prof. Dr. Kusworo Adi, MT.

III. EXPERIENCE

2020 – 2021	Designation: Research and Teaching Assistant in Prof. Wisnu Jatmiko Laboratory (https://cs.ui.ac.id/personnel/wisnu-jatmiko/) Faculty of Computer Science, Universitas Indonesia Instruction Areas: Assisting undergraduate courses including subjects such as
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Robotics, Machine Learning, Research Methodology, Computer and Society, Fundamental Physics, and Embedded Systems. Assisting research on image processing, video processing, robotics, simulation, and writing articles.

IV. AWARDS & ACHIEVEMENTS

- Nominated for Best Conference Paper Award of 2025 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), Melbourne, Australia.
- Best Conference Paper Award of 9th Seminar Nasional Sistem Informasi dan Teknologi (SISFOTEK) 2025, Politeknik Negeri Batam, Indonesia.
- Best Conference Paper Award of 2024 IEEE 10th International Conference on Applied System Innovation (IEEE ICASI 2024), Kyoto, Japan.
- Best Conference Paper Award of 6th IEEE International Conference on Knowledge Innovation and Invention (ICKII) 2023, Sapporo, Hokkaido, Japan.
- Best Session Presenter Award (Data Science and Artificial Intelligence Session) in ICACSIS (International Conference on Advanced Computer Science and Information Systems) 2022, Depok, Indonesia (Virtual Conference).
- Winner of Summer Web Campus Visit Student Video Contest by Kanazawa University 2022.
- Best Paper Award for AI, IOT and Computer Vision Track in IEEE TENSYP (Region Ten Symposium) 2020, Dhaka, Bangladesh (Virtual Conference).
- Universitas Diponegoro Cum Laude Graduate Award (2015).
- 2nd Winner of Writing Competition, Faculty of Science and Mathematics Essay Week, Universitas Diponegoro (2014).

V. INVITED SPEAKERS/ LECTURES

- International Seminar on New Paradigm and Innovation on Natural Sciences and its Application (ISNPINSA 2025), Title: “*AI-Driven Evolution in Unmanned Surface Vehicles: From Path Planning to Swarm Intelligence*” by Universitas Diponegoro, Semarang, Indonesia (16 October 2025).
- Guest Lecture Title: “*Introduction to Artificial Intelligence: What, Why, and How*”, by Department of Informatics, Universitas Diponegoro, Semarang, Indonesia, (22 May 2025).
- Guest Lecture Title: “*Artificial Intelligence Applications for Physics Research*”, by Department of Physics, Universitas Diponegoro, Semarang, Indonesia, (November 2021).

VI. RESEARCH

Publications:

- 2026 **Hamid, N.**, Antariksa, G., Dharmawan, W., Jati, G., Saleh, H., El, S., 2026. AI-driven swarm USV operations: A comprehensive bibliometric and analytical review. *Advanced Engineering Informatics* 69, 103972. <https://doi.org/10.1016/j.aei.2025.103972>
- 2025 Prayogo, R.D., **Hamid, N.** and Nambo, H., (2025). Hybrid CNN-Based Transfer Learning Enhances Brain Tumor Classification on MRI Images. *IEEE Access*. (13) 116654 – 116668 10.1109/ACCESS.2025.3584376
- 2024 Dwiyanti, L., Nambo, H., & **Hamid, N.** (2024). *Leveraging Explainable Artificial Intelligence (XAI) for Expert Interpretability in Predicting Rapid Kidney Enlargement Risks in Autosomal*

- Dominant Polycystic Kidney Disease (ADPKD)*. AI (Switzerland), 5(4), 2037–2065. <https://doi.org/10.3390/ai5040100>
- 2024 **Hamid, N.**, Rumianti, F., Prayogo, R. D., & Nambo, H. (2024). *Machine Learning and Deep Learning Approach for Predicting Neonatal Mortality in Indonesia*. 2024 10th International Conference on Applied System Innovation (ICASI), 1–3. <https://doi.org/10.1109/icas60819.2024.10547863>
- 2024 Prayogo, R. D., **Hamid, N.**, & Nambo, H. (2024). *An Improved Transfer Learning-Based Model with Data Augmentation for Brain Tumor Detection*. 2024 International Conference on Artificial Intelligence in Information and Communication (ICAIIIC), 602–607. <https://doi.org/10.1109/ICAIIIC60209.2024.10463471>
- 2023 **Hamid, N.**; Dharmawan, W.; Nambo, H. *Dynamic Path Planning for Unmanned Surface Vehicles with a Modified Neuronal Genetic Algorithm*. Applied System Innovation Journal. 2023, 6, 109. <https://doi.org/10.3390/asi6060109>
- 2023 **N. Hamid**, W. Dharmawan, and H. Nambo, “*Neural Network-based Genetic Algorithm for Autonomous Boat Pathfinding*,” in 2023 IEEE 6th International Conference on Knowledge Innovation and Invention (ICKII), IEEE, 2023, pp. 497–502. doi: 10.1109/ICKII58656.2023.10332606.
- 2022 Willy Dharmawan, Pam Thi Loc, **Nur Hamid**, Hidetaka Nambo, “*Protein-Ligand Pair Binding Prediction Using Wide Resnet for Virtual Drug Screening*”, in Proceeding of Asia Pacific Industrial Engineering & Management Systems Conference 2022, pp. 1-7.
- 2022 **Nur Hamid**, Willy Dharmawan, Hidetaka Nambo, “*Autonomous Evacuation Boat in Dynamic Flood Disaster Environment*,” in Proceedings - ICACSI 2022: 14th International Conference on Advanced Computer Science and Information Systems, IEEE, 2022, pp. 117–122. doi: 10.1109/ICACSI56558.2022.9923446.
- 2020 Ahmad Gamal, Ari Wibisono, Satrio Bagus Wicaksono, Muhammad Alvin Abyan, **Nur Hamid**, Hanif Arif Wisesa, Wisnu Jatmiko & Ronny Ardhianto, *Automatic LIDAR building segmentation based on DGCNN and euclidean clustering*. J Big Data 7, 102 (2020). <https://doi.org/10.1186/s40537-020-00374-x>
- 2020 **Nur Hamid**, Ari Wibisono, Ahmad Gamal, Ronni Ardhianto, and Wisnu Jatmiko, *Adaptive Update in Deep Learning Algorithms for LiDAR Data Semantic Segmentation*, IEEE Region 10 Symposium 2020, page 1038 – 1041, ISBN: 978-1-7281-7366-5.
- 2020 Noverina Alfiany, Grafika Jati, **Nur Hamid**, Rifat Ahdi Ramadhani, Made Wira Dhanar Santika and Wisnu Jatmiko, *Kinematics and Simulation Model of Autonomous Indonesian “Becak” Robot*, IEEE Region 10 Symposium 2020, page 1692 – 1695, ISBN: 978-1-7281-7366-5.
- 2019 **Nur Hamid**, Ari Wibisono, M. Anwar Ma’sum, Ahmad Gamal, Ronni Ardhianto, Aniati Murni Arymurthy, and Wisnu Jatmiko, *3D Edge Convolution in Deep Neural Network Implementation for Land Cover Semantic Segmentation of Airborne LiDAR Data*, ACIRS 2019, page 216 – 220, ISBN 978-1-7281-2229-8.
- 2019 Noverina Alfiany, **Nur Hamid**, Grafika Jati, M. Anwar Ma’sum, and Wisnu Jatmiko, *Kinematics and Dynamics Analysis of An Autonomous Three-Wheeled Bicycle Modeling*, ACIRS 2019, page 17 – 21, ISBN 978-1-7281-2229-8.

- 2018 **Nur Hamid**, and Wahyu Catur W., *Wind Speed Forecasting Using Multivariate Time-Series Radial Basis Function Neural Network*, ICACSIS 2018, page 423 – 428, ISBN: 978-1-7281-0135-4.
- 2015 **Nur Hamid**, and Kusworo A., *Penentuan Tingkat Buta Warna dengan Metode Segmentasi Ruang Warna Fuzzy dan Rule-based forward chaining pada Citra Ishihara*, Youngster Physics Journal Vol. 2 No. 4, April 2015, Hal 211-218, ISSN: 2302-7371.

Conference:

- 2025 International Seminar on New Paradigm and Innovation on Natural Sciences and its Application (ISNPINSA 2025), Universitas Diponegoro, Semarang, Indonesia (Invited Speaker).
- 2024 IEEE 10th International Conference on Applied System Innovation (IEEE ICASI 2024), Kyoto, Japan (1st author and presenter).
- 2023 The 6th IEEE International Conference on Knowledge Innovation and Invention 2023 (1st author and presenter).
- 2022 The 14th IEEE International Conference on Advanced Computer Science and Information Systems (ICACSIS) 2022 (1st author and presenter).
- 2020 IEEE Region 10 Symposium (TENSYP) 2020, Dhaka, Bangladesh (Fully Virtual Conference) (1st Author and Presenter)
- 2018 International Conference on Advanced Computer Science and Information Systems (ICACSIS) 2018 (1st author and presenter).

Experience as a reviewer of scientific articles:

- Reviewer at the Journal of Undergraduate Research International (JURI)
- Reviewer at the 2025 IEEE International Conference on Data and Software Engineering
- Reviewer at the Journal of IEEE Sensor Letters
- Reviewer at The Telematika Journal
- Reviewer at International Conference on Advanced Computer Science and Information Systems

VII. SCHOLARSHIP

- 2021 - 2024 MEXT Scholarship (Japan) and JICA Innovative Asia batch 5.
- 2018 - 2020 Indonesian Education Scholarship for Education Fund Management Institution (LPDP).
- 2015 TOEFL Scholarship from TEST English School.
- 2015 English Language Scholarship from Mata Air Foundation.
- 2011 -2015 Bidikmisi Scholarship from Ministry of Research, Technology and Higher Education.

VIII. INTERNSHIP, TRAINING, AND PARTICIPATION

Internship Program:

- Internship program at Eftax Co. Ltd., Scribble Osaka Lab, Osaka, Japan (16 August – 5 September 2022) homepage: <https://eftax.co.jp/en/>

Training Program:

- Training on the use of NVIDIA DGX-1, by the Faculty of Computer Science, Universitas Indonesia 2019.
- Public Speaking Training (PST), Al Hikam Depok Islamic Student Boarding School, 2018.
- Preparation for Departure Scholarship for Education Fund Management Institution (LPDP), Depok, November 2016.
- TOEFL & IELTS Class, TEST English School, Pare, Kediri, 2015.

Organizational Involvement Experience:

- Indonesian Representative at International Student Committee (ISC), Graduate School of Natural Science and Technology, Kanazawa University.
- Vice Chair of Management of the LPDP UI Scholarship Recipient Student Association for the 6.0-year period.
- Head of Communication and Information Sector - Postgraduate Muslim Student Association (HIMMPAS) Universitas Indonesia.
- Preparation for Departure (PK) Scholarship for Education Fund Management Institution (LPDP), November 2016 (Chairperson of Pre-PK).
- Electronics and Instrumentation Study Group, (Research and Development Staff).
- Chair of Physics Student Association (HMF) Diponegoro University 2013.

IX. REFERENCES**Prof. Dr. Sami El-Ferik** (Director of IRC-SML)

- Affiliation: Control and Instrumentation Engineering Department, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia
- Email: selferik@kfupm.edu.sa

Dr. Haitham Saleh (Postdoctoral supervisor)

- Affiliation: Industrial and Systems Engineering Department, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia
- Email: haitham.saleh@kfupm.edu.sa

Prof. Hidetaka Nambo (Doctoral degree supervisor)

- Affiliation: Graduate School of Natural Science and Technology, Kanazawa University, Japan
- Email: nambo@blitz.ec.t.kanazawa-u.ac.jp

Prof. Dr. Eng. Wisnu Jatmiko (master's degree supervisor)

- Affiliation: Faculty of Computer Science, Universitas Indonesia, Indonesia
- Email: wisnuj@cs.ui.ac.id

Prof. Dr. Kusworo Adi, MT. (bachelor's degree supervisor)

- Affiliation: Faculty of Mathematics and Natural Science, Universitas Diponegoro
- Email: kusworoadi@gmail.com